Name

Mathematics Advanced Year 11

Exponential Functions

Book 1

Version: 261021

Feedback:
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Syllabus Content

MAV-11-08 analyses graphs of exponential and logarithmic functions

Exponential functions

- Graph the exponential functions $y = k(a^x)$ and $y = k(a^{-x})$ for constants a and k where $a > 0$, $a \neq 1$ and $k \neq 0$, and identify its asymptote, y-intercept, domain and range
- Describe the behaviour of $y = k(a^x)$ and $y = k(a^{-x})$ as $x \to \infty$ and $x \to -\infty$
- Examine the gradient of the tangent to the curve $y = a^x$ at its y-intercept for varying values of a, and verify using graphing applications that there is a unique number $e \approx 2.71828182845...$, such that the gradient of the tangent to $y = e^x$ at $x = 0$ is 1, and call this number e Euler's number
- Examine the gradient function of $y = a^x$ with graphing applications and identify that the gradient function is $\frac{d}{dx}(a^x) = ka^x$, for some constant k depending only on a
- Conclude that Euler's number, e, is a unique number such that $\frac{d}{dx}(e^x) = e^x$, that is, e^x is its own derivative

Properties of the Exponential Curve

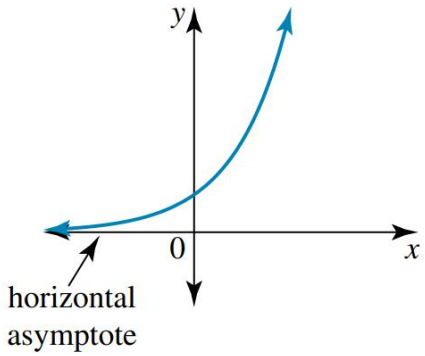
ⓘ Exponential Curves

An **exponential relationship** $y = a^x$ produces **exponential curve** graphs.
Exponential curves have these features:

- An **asymptote**: a line the graph approaches but never reaches.
- The **nature** of the graph (**growth** or **decay**) depends on the sign of x.
- Consecutive y values change by a **common ratio**, which is the base number a.

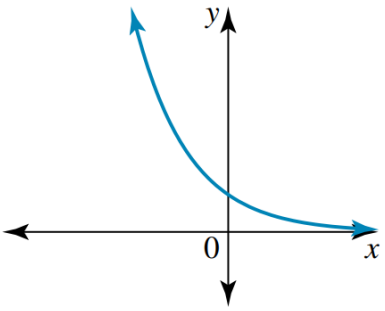
$y = a^x, \quad a > 1$

Increasing, exponential growth



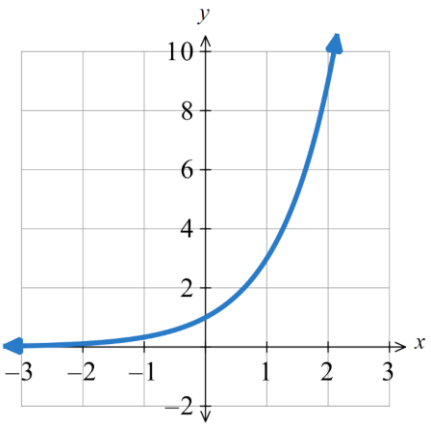
$y = a^{-x}, \quad a > 1$

Decreasing, exponential decay



ⓘ **Identify** key features of an exponential curve

a



y – intercept:

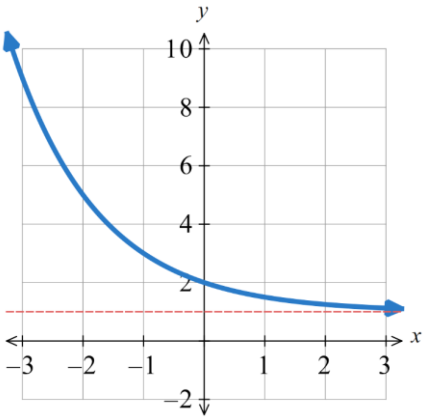
Asymptote:

Nature:

What happens to y when $x \to \infty$?

What happens to y when $x \to -\infty$?

b



y – intercept:

Asymptote:

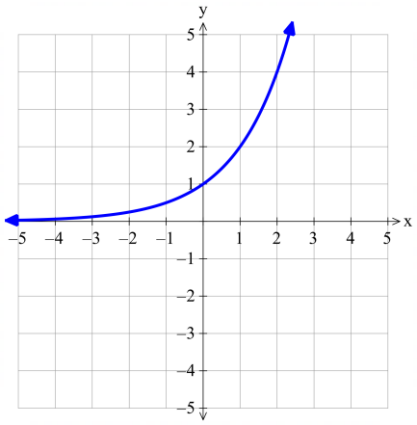
Nature:

What happens to y when $x \to \infty$?

What happens to y when $x \to -\infty$?

1 Consider these exponential curves.

a $y = 2^x$



y-intercept:

$(0,1)$

Asymptote:

$y = 0$

Nature:

Increasing; exponential growth

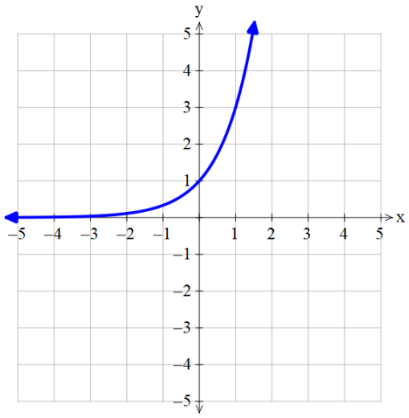
What happens when $x \to \infty$?

$y \to \infty$

What happens when $x \to -\infty$?

$y \to 0$

b $y = 3^x$



y-intercept:

$(0,1)$

Asymptote:

$y = 0$

Nature:

Increasing; exponential growth

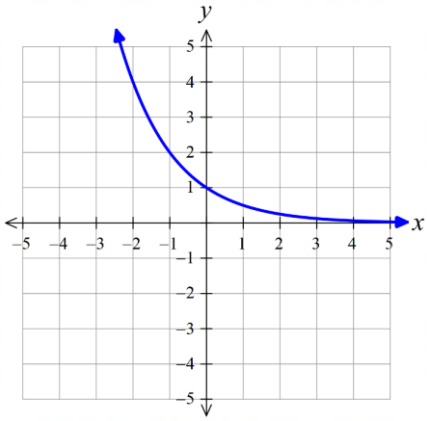
What happens when $x \to \infty$?

$y \to \infty$

What happens when $x \to -\infty$?

$y \to 0$

c $y = \left(\frac{1}{2}\right)^x$



y-intercept:

$(0,1)$

Asymptote:

$y = 0$

Nature:

decreasing; exponential decay

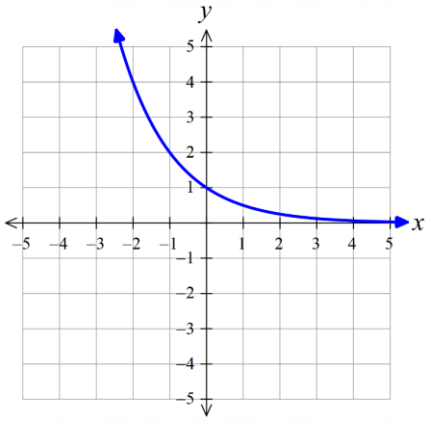
What happens when $x \to \infty$?

$y \to 0$

What happens when $x \to -\infty$?

$y \to \infty$

d $y = 2^{-x}$



y-intercept:

$(0,1)$

Asymptote:

$y = 0$

Nature:

decreasing; exponential decay

What happens when $x \to \infty$?

$y \to 0$

What happens when $x \to -\infty$?

$y \to \infty$

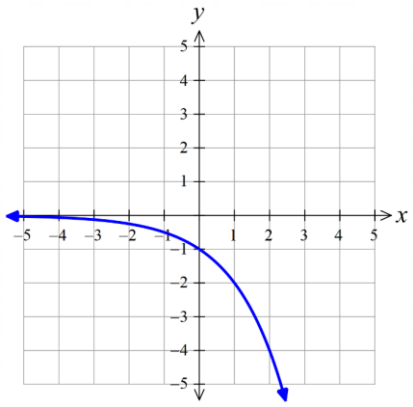
2 What do you notice about part **c** and **d** above?

.....

They are the same.

3 Consider these exponential curves.

a $y = -2^x$



y-intercept:

$(0, -1)$

Asymptote:

$y = 0$

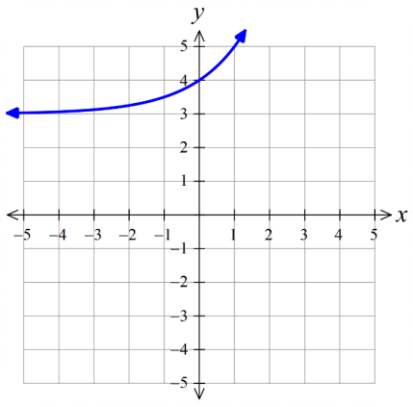
What happens when $x \to \infty$?

$y \to -\infty$

What happens when $x \to -\infty$?

$y \to 0$

b $y = 2^x + 3$



y-intercept:

$(0,4)$

Asymptote:

$y = 3$

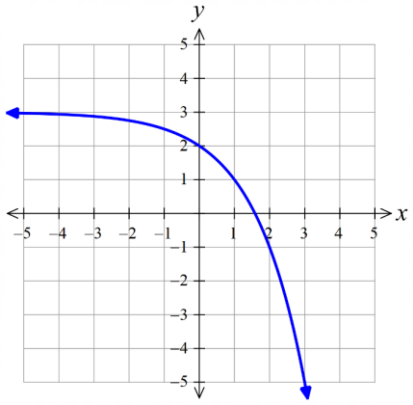
What happens when $x \to \infty$?

$y \to \infty$

What happens when $x \to -\infty$?

$y \to 3$

c $y = -2^x + 3$



y-intercept:

$(0,2)$

Asymptote:

$y = 3$

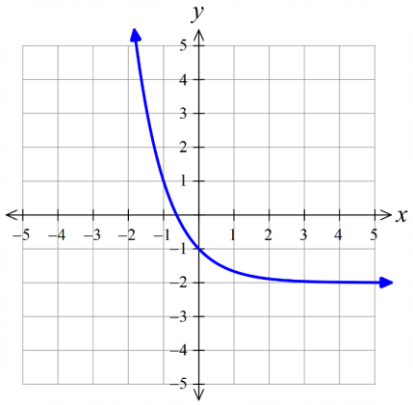
What happens when $x \to \infty$?

$y \to -\infty$

What happens when $x \to -\infty$?

$y \to 3$

d $y = \left(\frac{1}{3}\right)^x - 2$



y-intercept:

$(0, -1)$

Asymptote:

$y = -2$

What happens when $x \to \infty$?

$y \to -2$

What happens when $x \to -\infty$?

$y \to -\infty$

Exponential Curve Transformations

ⓘ Exponential Curve Transformations

$a > 1$ exponential growth
$0 < a < 1$ exponential decay
Common ratio

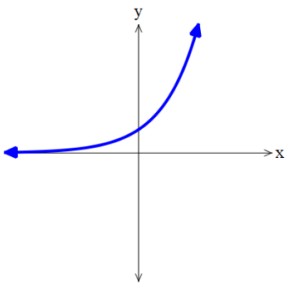
$$y = k \cdot a^x$$

Factor; affects y-intercept and reflects across x axis if negative

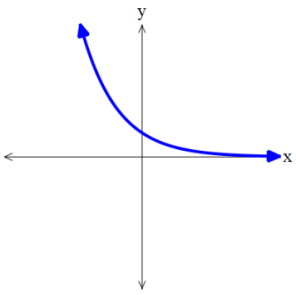
$a > 1$ exponential decay
$0 < a < 1$ exponential growth
Common ratio

$$y = k \cdot a^{-x}$$

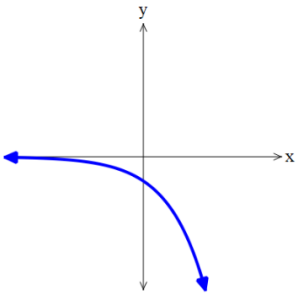
$y = 2^x$



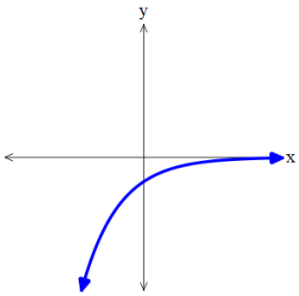
$y = 2^{-x}$ or $y = \left(\frac{1}{2}\right)^x$



$y = -2^x$



$y = -2^{-x}$ or $y = -\left(\frac{1}{2}\right)^x$



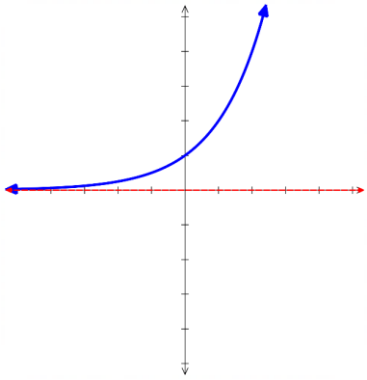
ⓘ Exponential Curve Transformations

Consider the equation of an exponential curve $y = k(a)^x + c$, where $a > 0$

The parameters a, c, k are numbers that affect the graph in predictable ways.

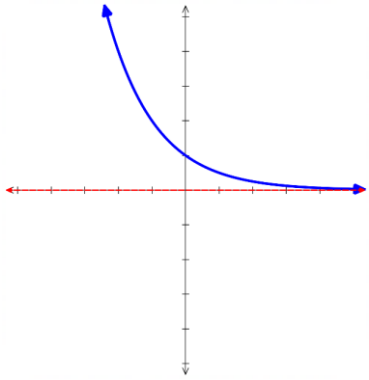
$a > 1$, we call this exponential ……

$y = 2^x$



$0 < a < 1$, we call this exponential ……

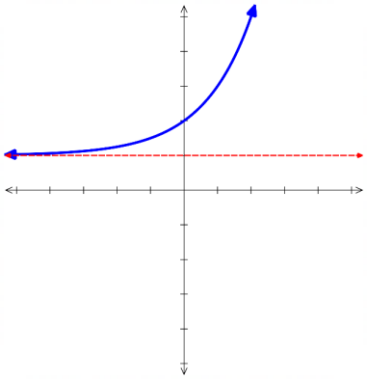
$y = \left(\frac{1}{2}\right)^x$



$c > 0$, the curve is shifted ……

The value of c is the ……

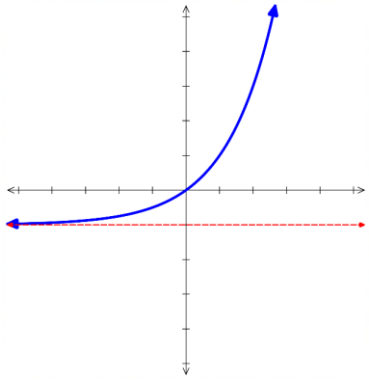
$y = 2^x + 1$



$c < 0$, the curve is shifted ……

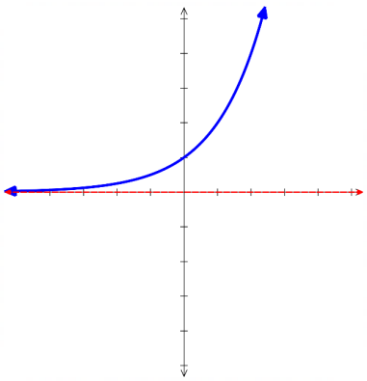
The value of c is the ……

$y = 2^x - 1$



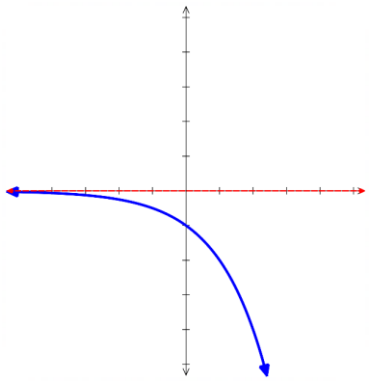
$k > 0$

$y = 1(2^x)$



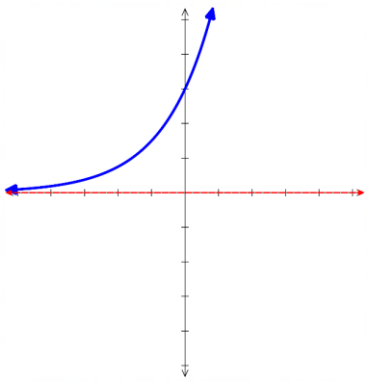
$k < 0$, reflect the graph across x-axis

$y = -1\ (2^x)$



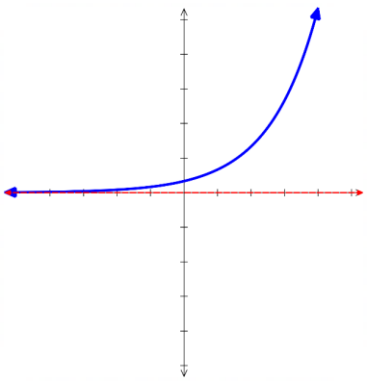
$|k| > 1$, the curve changes more quickly.

$y = 3(2^x)$



$0 < |k| < 1$ the curve changes more slowly.

$y = \frac{1}{3}(2^x)$



ⓘ Exponential Curve Transformations

$a > 1$ exponential growth
$0 < a < 1$ exponential decay
Common ratio

$$y = k \cdot a^x + c$$

Factor; affects y-intercept and reflects across x axis if negative

vertical translation $y = c$ is asymptote

$a > 1$ exponential decay
$0 < a < 1$ exponential growth
Common ratio

$$y = k \cdot a^{-x} + c$$

To sketch exponential curves:

1. Find y-intercept by letting $x = 0$.
2. Find asymptote: $y = c$.
3. Plot at least one other point, such as at $x = 1$.

⚠ You do not need to find x-intercepts until you learn logarithms

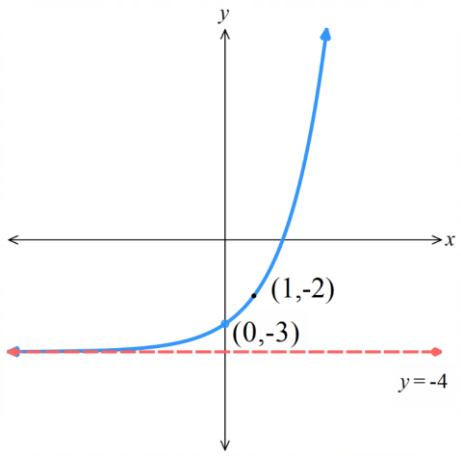
Example Sketch transformations of exponential curves

$y = 2^x - 4$

1. y-int: Let $x = 0$, $y = 2^0 - 4 = -3$

2. Asymptote: $y = -4$

3. When $x = 1$: $y = 2^1 - 4$, $y = -2$

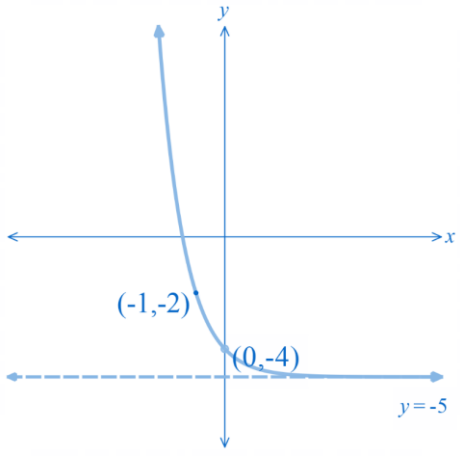


$y = 3^{-x} - 5$

1. y-int:

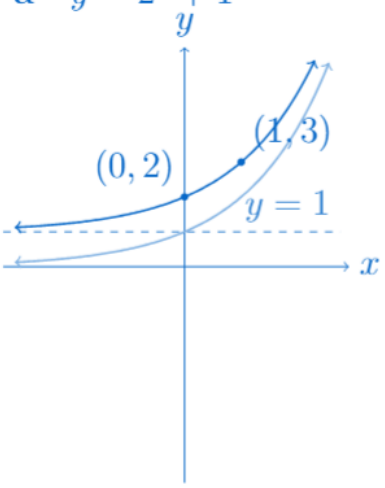
2. Asymptote:

3. Another point:

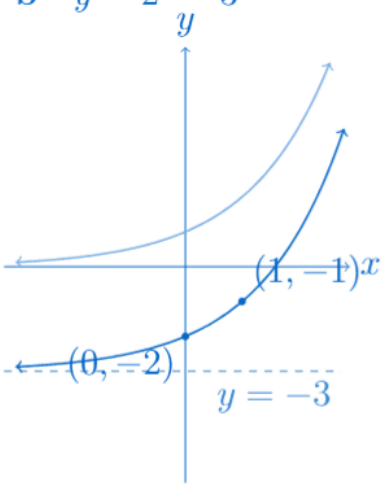


1 The graph of $y = 2^x$ is given. Sketch these graphs on the same axis to show the transformation. Clearly label the y-intercept, asymptote, and the point where $x = 1$.

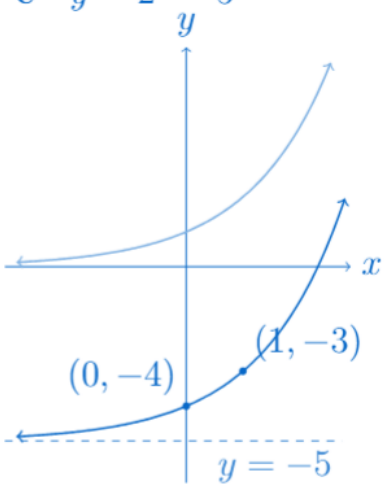
a $y = 2^x + 1$



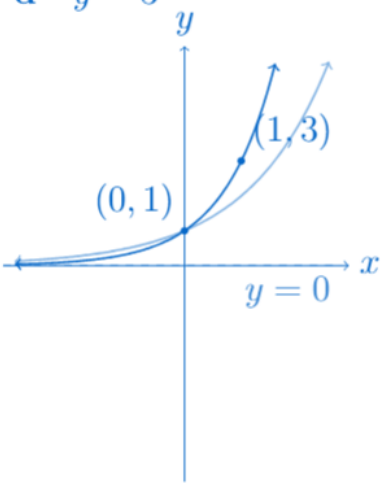
b $y = 2^x - 3$



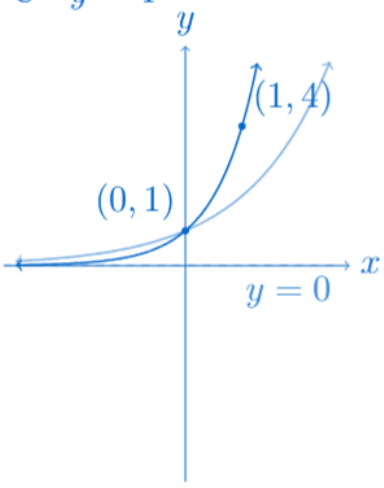
c $y = 2^x - 5$



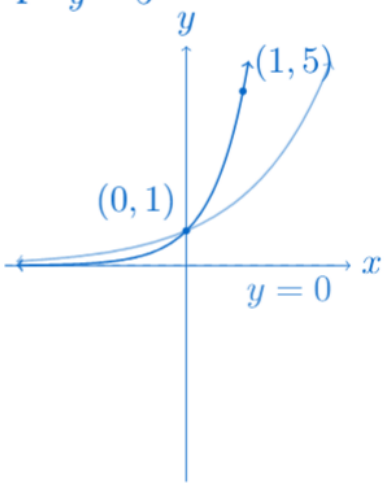
d $y = 3^x$



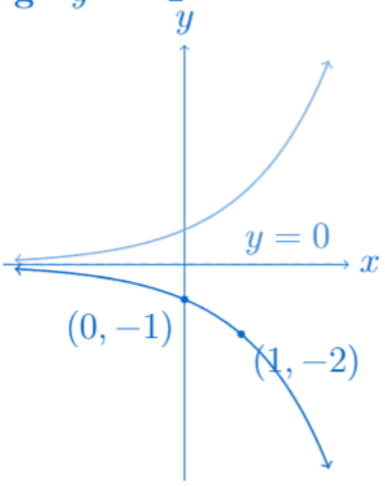
e $y = 4^x$



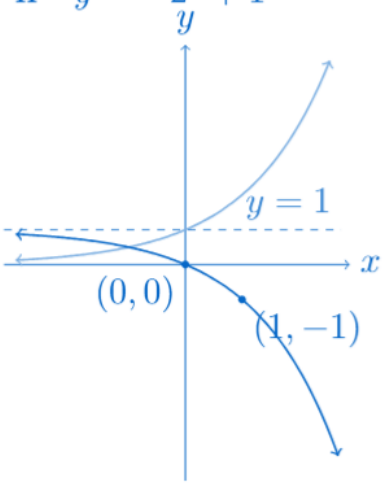
f $y = 5^x$



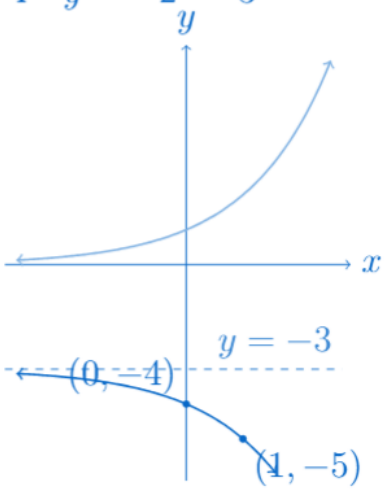
g $y = -2^x$



h $y = -2^x + 1$



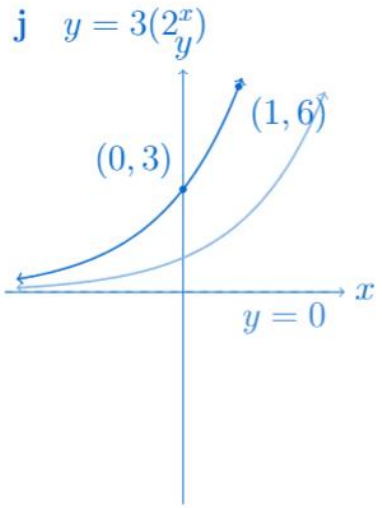
i $y = -2^x - 3$

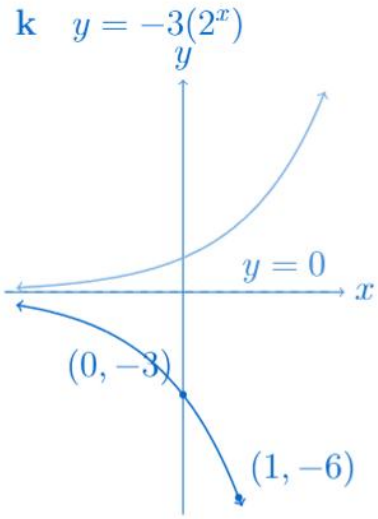


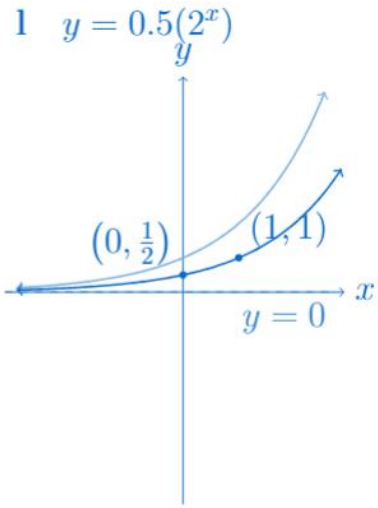
j $y = 3(2^x)$

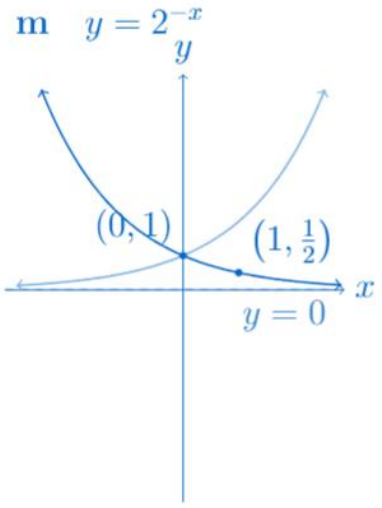
k $y = -3(2^x)$

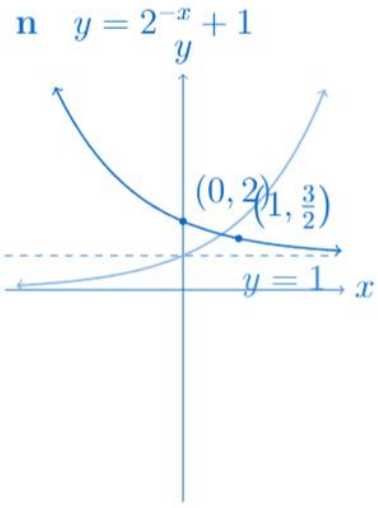
l $y = 0.5(2^x)$

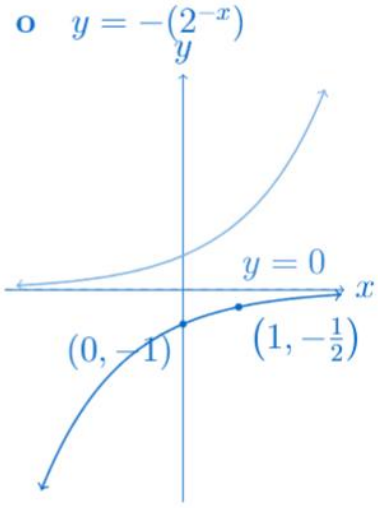


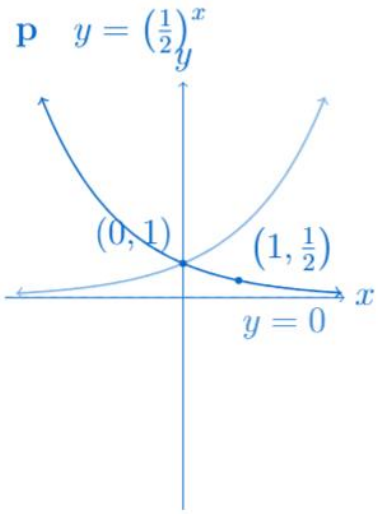


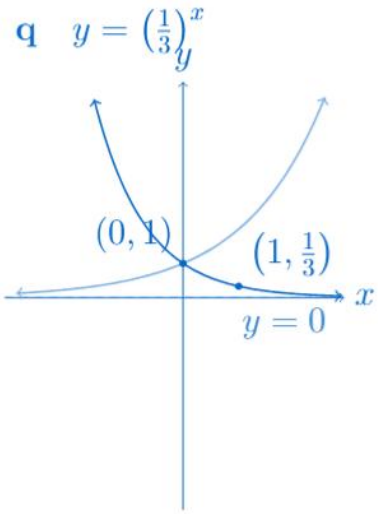


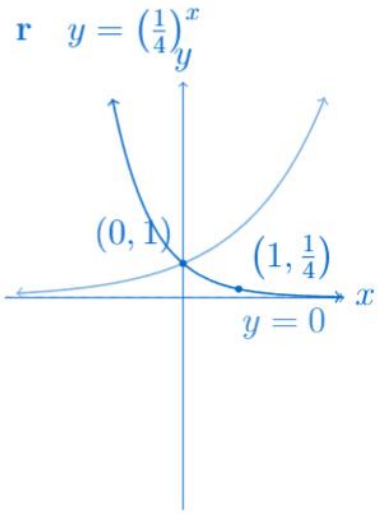






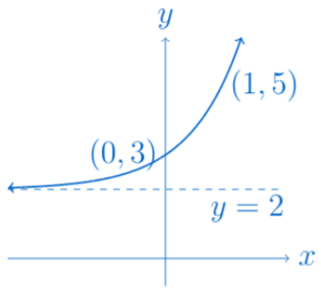




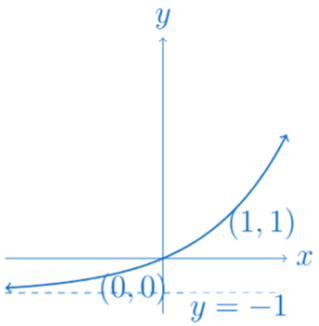


2 Sketch these exponential equations.
Clearly label the y-intercept, asymptote, and the point where $x = 1$.

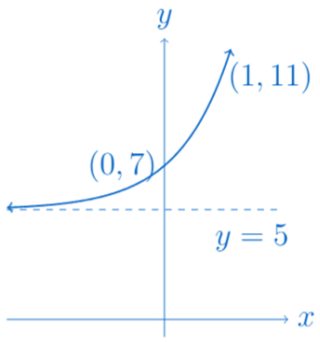
a $y = 3^x + 2$



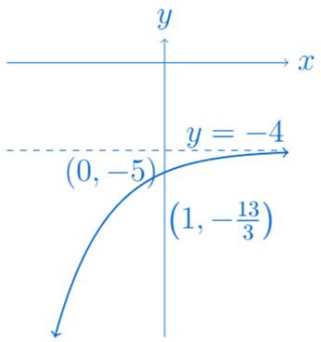
b $y = 2^x - 1$



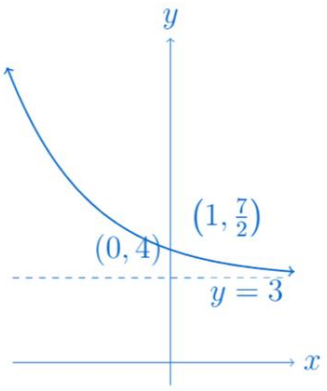
c $y = 2(3^x) + 5$



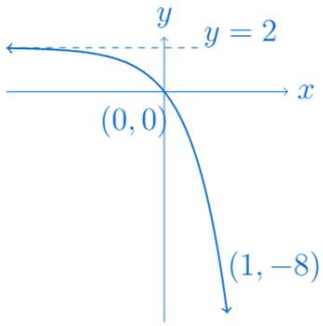
d $y = -3^{-x} - 4$



e $y = 2^{-x} + 3$



f $y = -2(5^x) + 2$

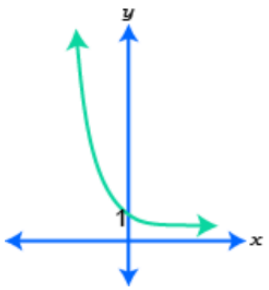


3 Match the equation to the graph.

$y = 6^x$ $y = -6^x$ $y = -6^x - 2$ $y = -6^x + 2$ $y = 6^{-x}$

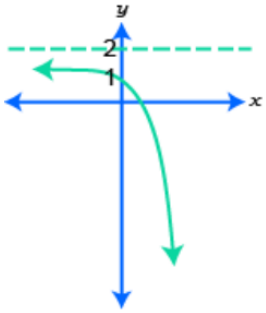
$y = 6^{-x} - 2$ $y = 6^{-x} + 2$ $y = -6^{-x}$ $y = -6^{-x} - 2$ $y = 6^{-x} + 2$

a



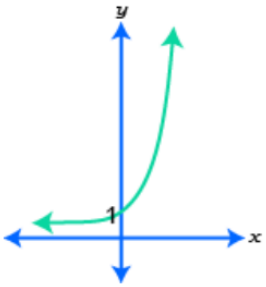
$y = 6^{-x}$

b



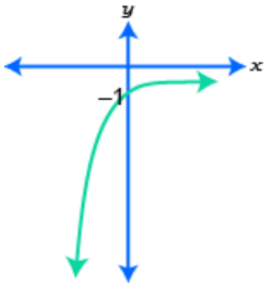
$y = -6^x + 2$

c



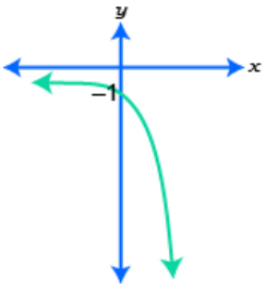
$y = 6^x$

d



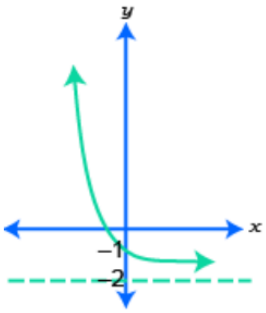
$y = -6^{-x}$

e



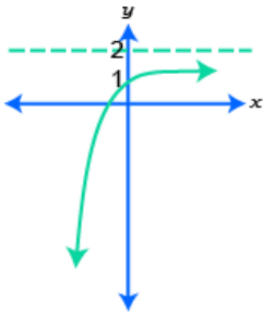
$y = -6^x$

f



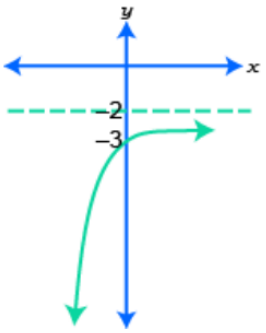
$y = 6^{-x} - 2$

g



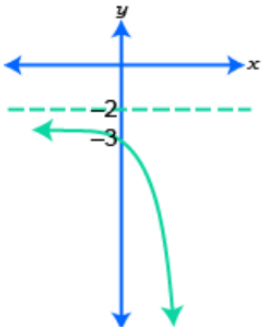
$y = -6^{-x} + 2$

h



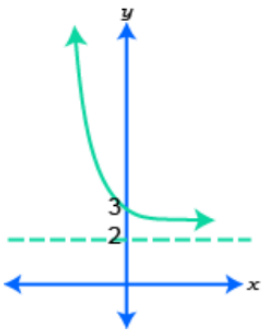
$y = -6^{-x} - 2$

i



$y = -6^x - 2$

j



$y = 6^{-x} + 2$

The Natural Exponential e

Investigation Euler's number e

Using Desmos:

1 Sketch the function $f(x) = 2^x$, then sketch $f'(x)$ on the same axis.
What do you notice about the shape of the derivative function?

2 Sketch the function $f(x) = 4^x$, then sketch $f'(x)$ on the same axis.
What do you notice about the shape of the derivative function?

3 For $f(x) = a^x$, when $a = 2$, $f'(x)$ is ..... than the original function.

when $a = 4$, $f'(x)$ is ..... than the original function.

So, we can write the derivative of $\frac{d}{dx}(a^x) =$ ..... a^x

So, there must be a value of a such that $f(x) = f'(x)$.

Use Desmos to sketch $f(x) = a^x$ and $f'(x)$.

Use the slider to find a value where the derivative function matches the original function.

What value of a is this?

$a =$ .....

We call this number .....'s number.

We call any exponentials in the form $f(x) =$ ..... "natural" exponentials.

ⓘ Natural Exponentials

Euler's number $e = 2.71828182845\ldots$ has a unique property:

$$\text{if } f(x) = e^x$$
$$\text{then } f'(x) = e^x$$

This means $f(x) = f'(x)$, the function equals its own derivative.

The rate of growth stays proportional to the current amount.

This exponential relationship models many real-life phenomena, such as:

population growth, disease spread, radioactive decay, compound interest.

FOUNDATION

1 Using a calculator, evaluate to 3 s.f.

a e^2

7.39

b $-\sqrt{e}$

−1.65

c $\frac{1}{e^3}$

0.0498

2 The mass of a radioactive substance in grams is given by $M = 150e^{-0.014t}$ where t is in years. Find these masses. Round to 2 d.p. where appropriate.

a Initial mass

150 g

b Mass after 10 years

130.40 g

c Mass after 250 years

4.53 g

d What will the mass become as $t \to \infty$ years?

Approaches 0 g

3 The population of koalas is currently increasing according to the formula $N = 1250e^{0.25t}$, where t is in months. Find these populations, rounding to the nearest koala.

a Initial population

1250

b Population after 12 months

25 107

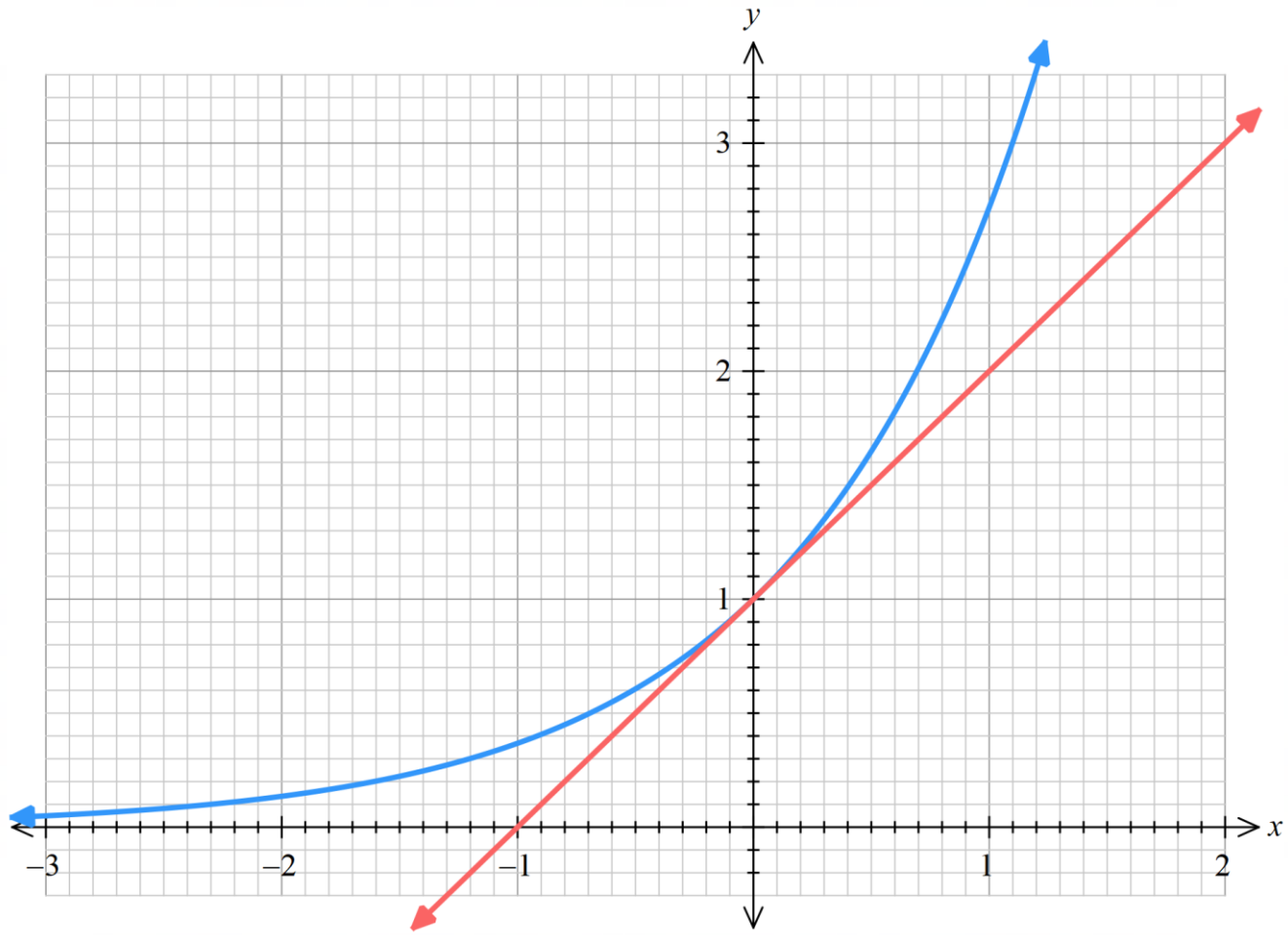
c Population after 5 years

4 086 271 716

d Is this a realistic model for the population of koalas long-term?

No, exponential growth cannot continue forever.

4 Consider the graph of $y = e^x$ shown below and its tangent line at $y = 1$.



a The tangent at $y = 1$ is shown on the graph.
By inspecting the line, what is the gradient of the tangent at $y = 1$?

1

b By drawing additional tangent lines at $y = \frac{1}{2}, y = 2, y = 3$, complete the table:

Function value y	$\frac{1}{2}$	1	2	3
Gradient $\frac{dy}{dx}$	$\frac{1}{2}$	1	2	3

c How is the gradient of the tangent related to the function value for $y = e^x$?

$\frac{dy}{dx} = y$
or, $f(x) = f'(x)$

d On the same graph as above, sketch tangent lines at $x = -1, x = 0, x = 1$.
What do you notice about the x-intercepts of the tangent lines?

The x intercepts are always 1 unit to the left of the point of tangency.